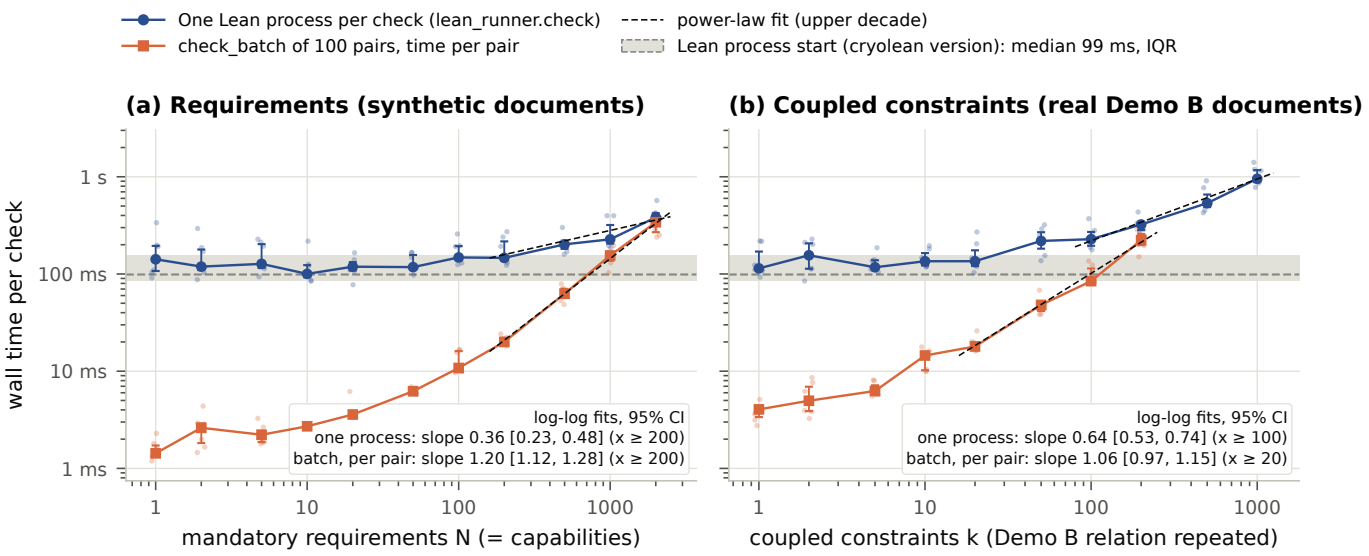
One Lean process per check (lean_runner.check)
check_batch of 100 pairs, time per pair
power-law fit (upper decade)
Lean process start (cryolean version): median 99 ms, IQR
(a) Requirements (synthetic documents)
(b) Coupled constraints (real Demo B documents)
wall time per check
1 s
100 ms
10 ms
1 ms
log-log fits, 95% CI
one process: slope 0.36 [0.23, 0.48] (x ≥ 200)
batch, per pair: slope 1.20 [1.12, 1.28] (x ≥ 200)
log-log fits, 95% CI
one process: slope 0.64 [0.53, 0.74] (x ≥ 100)
batch, per pair: slope 1.06 [0.97, 1.15] (x ≥ 20)
1
10
100
1000
mandatory requirements N (= capabilities)
coupled constraints k (Demo B relation repeated)